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FDM 4D printing: A low-cost approach of shape programming and assessing the shape memory properties using angle measurement methods in hot water actuation testing apparatus

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ABSTRACT

With the advent of four-dimensional (4D) printing technology, manufactured components may adapt their form, size, or properties in response to external stimuli. This new 4D printing technology has opened several doors in the science, technology, engineering, and mathematics (STEM) disciplines. Fused deposition modeling (FDM) is one of the most common and cost-effective 3D printing procedures for making components out of shape-memory polymers. It is well known that programming and training are required for the 4D printing process. This work reports crucial programming methodologies that can be applied to FDM 4D printing. A detailed angle measurement scheme and the hot water actuation apparatus have been reported and discussed. This can be used to measure shape morphing in a folding or bending actuator for a variety of multifaceted applications. The angle measurement scheme presented allows for the accurate measurement of the degree of bending without causing any damage to the actuator and for real-time feedback regarding how much shape-morphing has taken place. In addition, the hot water actuation apparatus can provide high accuracy in controlling the bending angle with a simple setup. This apparatus provides a reliable and cost-effective solution to obtaining accurate bending angle measurements. Finally, when it comes to accurately measuring the shape morphing in a bending actuator, the combination of the various programming methodologies, angle measurement scheme, and hot water actuation apparatus offers a number of advantages.
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1. Introduction

Additive manufacturing (AM) has opened up new ways to design products, making it possible to make things with complex internal structures that couldn't be made with old methods. By utilizing the advantages of AM, such as cost-efficiency, waste reduction, and rapid manufacturing, it is now possible to produce nature-inspired structures with high precision and accuracy using AM technology, such as lattices, trusses, honeycomb structures, and foldable designs. Four-dimensional (4D) printing has received widespread attention since its conceptualization in 2013 [1] due to its promise of revolutionizing product design and engineering. 4D printing is a method of manufacturing that uses a three-dimensional (3D) printer to create 3D objects from digital designs using smart materials, with the fourth dimension being time. Additionally, the smart materials used in 4D printing can respond to changes in their environment; this dynamic property enables objects with dynamic functionalities that would be difficult to create using other methods. This has allowed for revolutionary applications in product design, from medical implants that can respond to physiological changes in the body [2] to aerospace components with smart materials that self-assemble at varying temperatures and pressures [3].

The evolution of this technology has allowed for a simple classification of 4D printing technology into three distinct categories presented in Fig. 1. The first is smart material-infused 3D printing, which improves the strength and characteristics of 3D-printed products at a low cost [45]. Using a hobbyist 3D printer, parts with

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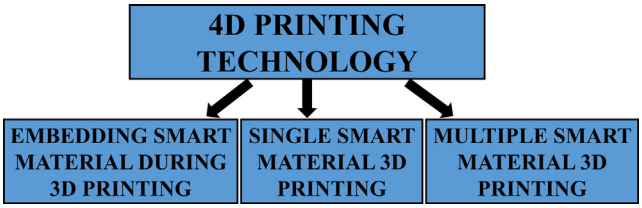


Fig. 1. Classification of 4D printing technology.

embedded, pre-programmed smart materials can be printed. Recently researchers have embedded a 4D printed conductive PLA inside the magnetorheological elastomers based on silicon and strontium ferrite to create lightweight and bistable composite actuators [6]. Researchers have also synthesized electrothermally responsive silicon/ethanol/NiCr composite for application in soft parallel manipulators [7]. A deep understanding of geometry for 4D printing and the precise location of smart material to be used is essential for this method. In the second category, direct 4D printing of smart material has been explored. It involves a novel 3D printing device that can directly print an intelligent material. Researchers have widely employed this method to create various 4D printed components for multiple applications and have also studied the effect of process parameters on shape memory behavior [8]. Printing shape memory polymers (SMPs), hydrogels, and tissue-printed biomedical implants are among its most well-known applications since their use is not restricted to a small number of studies. Researchers have utilized this method to study cold, warm, and hot programming of poly-ethylene terephthalate glycol (PETG) and have shown that the programming temperature is an important parameter for FDM 4D printing [9]. The third kind of 3D printing uses several materials, one or both of which may be smart materials. Multi-material 3D printers have advanced quickly in recent years, making this the greatest choice. However, it can only be applied to selected materials, particularly the same substance as the second categorization.

Among various smart materials and 3D printing methods, one of the most widely accepted and economical methods of 4D printing is the processing of thermo-responsive shape memory polymer (TR-SMP) by fused filament fabrication (FFF) because of the scalability, low cost, easy handling, and availability of a wide range of materials [10]. Similarly, SMPs have shown to be particularly helpful because of their low production costs, rapid recovery times, and extensive property tuning options. Programming or training is an important aspect of smart materials; therefore, there is a need to form different programming methodologies for 4D printing. 4D printing differs from 3D printing primarily because it necessitates adding programmed time-based alterations to 3D manufactured structures. The selection of smart materials, programming strategies, printing process, and stimulus-induced reactions are all essential for 4D printing [11]. Researchers have utilized various techniques for controlling the deformation of a bilayer actuator in 4D printing with temperature. The bilayer actuator's thermo-mechanical mechanism is examined, and three types of deformation behavior are modeled [12]. One of the most important challenges of controlling the shape morphing behavior of the 4D printed part is successfully solved by the researchers by varying the patterns and thickness of the chitosan/carbon black based polymer composite printed using direct ink writing (DIW) on polystyrene (PS) and further actuated by photothermal effect [13]. A novel approach to predict and control the shape-changing behavior for large-scale 4D printing of composite using an automated fiber placement machine is presented by using the technique of incremental change in coordinate axes after each increment methodology [14]. The Researchers examine the self-folding mechanism to understand how printing speed and liquefier temperature influence the layer-by-layer programming process and subsequent shape change. Additive fabrication with functional programming allows for what is known as functionally graded 4D printing [15]. SMPs are a kind of multifunctional smart materials with the ability to revert to their original shape using a cold/ hot programming technique. The programming environment's temperature has an important role. Researchers have studied that SMP-based energy absorbers undergo a cold-programming process at ambient temperature, which is lower than their glassy transition temperature [16]. In 4D printing, the requirement for thermo-mechanical programming can be reduced by manipulating the printing settings to generate internal strains into the actuators. It might be a helpful technique for achieving a precise shape memory effect. It is possible to regulate the amount of deflection in the actuators by using variable-stiffness patterning. Large deformations in the actuators are generated by the internal strain between the layers using patterns of varying sizes. This strain is highly tunable in terms of its magnitude and application direction [17].

Recently, researchers have tried to solve the problem of understanding the very difficult concept of 4D bioprinting using the DIW method through an easy-to-implement and follow learning module that can be used for teaching 4D bioprinting in class. It can be greatly useful for the development of 4D printing skills in early career researchers and students to improve the knowledge base of this novel technology [18]. For further widespread use of 4D printing technology, low-cost devices, programming methods, and characterization techniques are much needed. The present research focuses on exploring various shape, and material programming strategies that can be applied to the FDM/ FFF 4D printed parts using the lowest possible resources. A low-cost robust hot water actuation testing apparatus for shape memory property characterization has been presented. An easy method of testing 4D printed samples through different angle measurement methodologies for folding/ bending actuators is also shown in detail.

2. Shape programming methodologies

2.1. Shape memory effect in TR-SMP

If the material's external environment changes, smart materials may adapt by changing their properties or shapes. The term thermo-responsive shape memory effect (TR-SME) refers to a smart material whose properties alter as a function of temperature. By monitoring their surroundings, smart materials may adapt their properties or shapes to meet changing conditions. The TR-SME of an SMP is not often a characteristic of the polymer itself. It is generally the result of the polymer's morphology, programming strategy, and manufacturing method. Any thermo-responsive SMP will need a programming and recovery phase. The SMP undergoes a series of processes that allow it to memorize the shape and revert to the remembered shape when a stimulus in the form of heat at a particular temperature is provided. The mechanism of the shape memory effect in any thermo-responsive SMP is due to any one of the three, i.e., (I) dual-state mechanism (DSM), (II) dual-component mechanism (DCM), and (III) partial-transition mechanism (PTM) as shown in Fig. 2 [19]. Materials with a DSM are pliable above their glass transition temperature (T_g) but harden significantly upon cooling (e.g., silicon). In DSM (Fig. 2(I)), polymers having T_g less than 0 degrees C will be rubbery at room temperature. Therefore it can be formed to some deformed shape, and then it is maintained when cooled below its T_g as the micro-Brownian motion is fixed in the glassy state. Heating back to Tg, the polymer will return back to a rubbery state as the micro-Brownian movement is reactivated [19]. DCM corresponds to

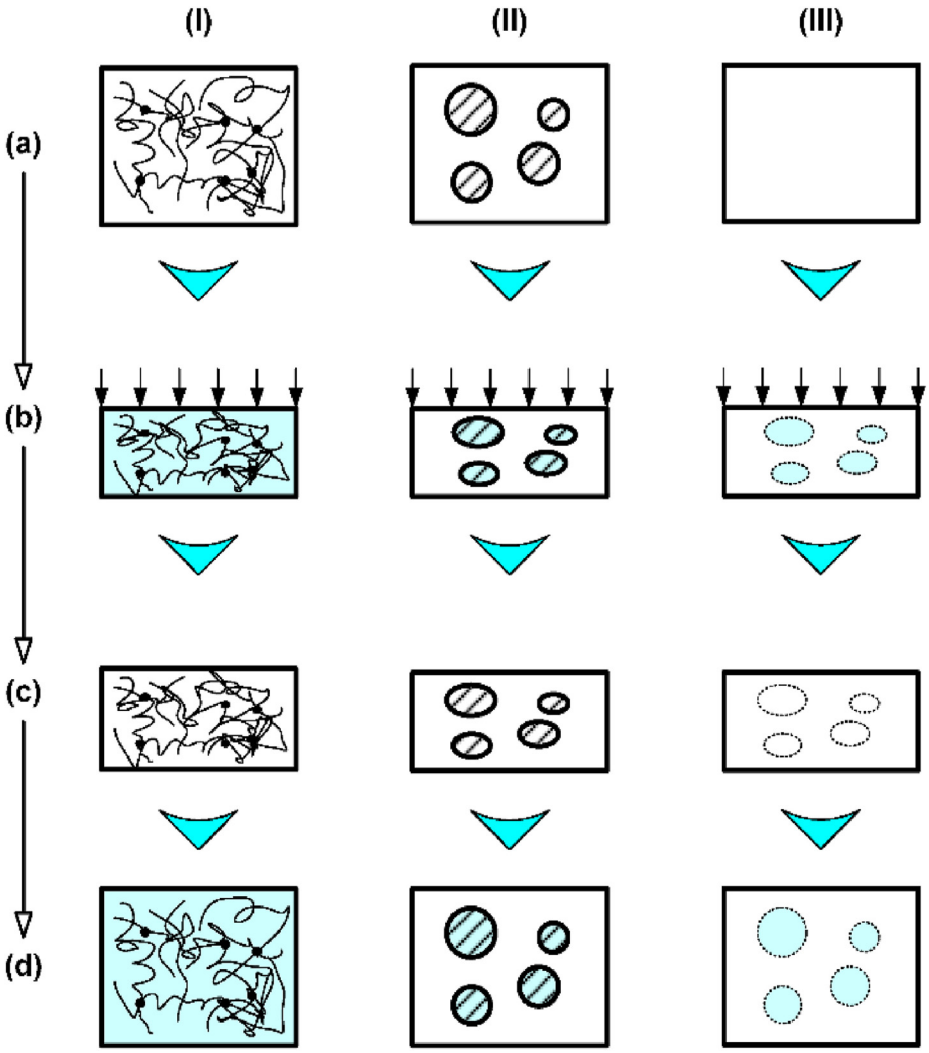


Fig. 2. Various mechanisms of shape memory effect in TR-SMP [19]: (I) DSM, (II) DCM, and (III) PTM. (a) shows the original sample at low temperatures, (b) is the sample upon heating and compressing, (c) is the sample after cooling and constraint removal, and (d) is the sample after heating for shape recovery.

materials with a soft and hard segmental structure in which each block can respond to temperature individually, e.g., polyurethane (PU). DCM mechanism is based on the variation of conformal entropy (Fig. 2(II)). At the start of the programming cycle, that is, while the temperature is low, the conformational entropy is high. When the temperature increases over the T_g during the deformation phase, the chain mobility increases, and the polymer softens. After that, entanglement/stretching occurs, and the entropy diminishes with cooling. When the distorted sample is reheated during the recovery phase, the polymeric chain mobility rises again, due to which the polymer returns to its original form by releasing the accumulated entropy and reverting to a thermodynamically preferred state, demonstrating the shape memory effect (SME) [20]. PTM (Fig. 2(III)) refers to materials where heating may stop at a temperature within the transition range instead of a complete transition in the DSM and DCM [21]. The heating may remain incomplete and completely stop in between before reaching the Tg. As a result, the unsoftened section of the polymer may store the elastic energy, and the softened part may behave as the transition component, thereby showing the shape memory effect.

The SMPs may be classified into three groups, as shown in Fig. 3 [22]: one-way shape memory effect (OWSME), two-way shape memory effect (TWSME), and multiple shape memory effect (multi-SME). Since the shape memory process can only go in one direction, the SMP must be artificially locked into a predetermined configuration before the procedure can begin again. OWSME describes how well these SMPs recover the temporary shape, and it is impossible to reverse the shape-changing effect by applying the same or different stimulus. In this case, there is only ever one direction of form change. A fresh round of programming is required each time the temporary form is desired (after the recovery process). TWSME is characterized by the reversibility in shape change, thereby remembering two temporary shapes. It is called a triple or multi-SME if the SMP can store information about three or more transient forms. With multi-SME, the distorted material gradually regains its original shape by transitioning between two or more temporary shapes [23].

2.2. Programming of TR-SMP in FDM 4D printing

Any thermo-responsive shape memory polymer requires a thermo-mechanical shape and material programming to show the shape memory effect in a pre-defined way. In the case of FDM 4D printing, the programming can be classified into three. The first one depends upon the pre-strain developed during the FDM process; therefore, various printing process parameters can be changed to show the controlled shape memory. The second programming can be realized by doing a set of post-process methods

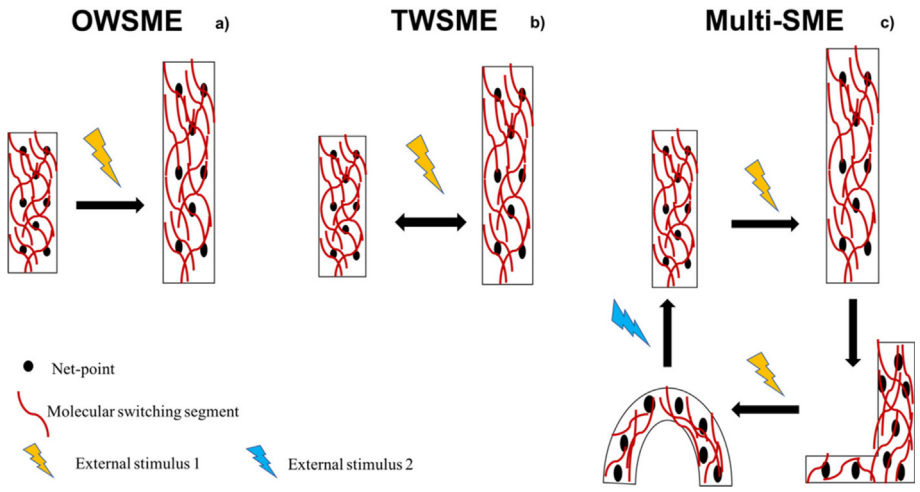


Fig. 3. SMPs with different shape-memory effects (SME) [22]: (a) one-way, (b) two-way, (c) multiple-SME.

like loading-annealing-cooling cycles. The third one is programming by variation material composition. If considered a mono-material FDM/ FFF 4D printing, then the first two types of programming are the most important.

2.2.1. Programming due to pre-strain during the FDM process

FDM technique, unlike many other types of 3D printing, uses a thermo-mechanical process on the material during the printing. An outline of the FDM procedure is shown in Fig. 4. Inside the liquefier, the material is heated to a higher temperature called programming temperature (T_p) than its glass transition temperature (T_g). The printer's head then moves at speed to deposit it onto the platform. This method produces a pre-strain by stretching the material at a high temperature, much like the heating-loading procedure for the SMP (heating and loading). Each printed layer is then cooled and hardened after being placed on the platform. This phase is analogous to the cooling phase in SMP [24]. After a layer is produced, the printing head moves on to the next layer, and the build platform moves lower. At last, the 3D-printed item is taken off the platform, marking the end of the thermo-mechanical programming process (mechanical unloading). Also, the initial layer is reheated since the hot material is put on top of it during the second printing pass. By reheating, the first layer to a certain temperature, some of the pre-strain may be relieved. When further layers are printed on top of the base layers, the base layers are continually warmed, releasing some of the pre-strains that had been stored in them. Since the nozzle leaves the last layer at the end of the 3D printing process, the final layer has the most pre-strain.

2.2.2. Programming by loading-annealing-quenching post-process

Programming by loading-annealing-quenching can be applied to the FDM 4D printed samples as a method of post-process after the completion of the printing, as shown in Fig. 5. The process entails using a fixture to set the SMP in the required form. The second step involves heating the sample to a temperature above the glass transition temperature (T_g), also known as the programming temperature (T_p), for a certain amount of time, often about 10 min. Then, the sample is cooled or quenched to room temperature. This completes the programming process, and the SMP has remembered the shape. Next comes the recovery phase, which entails applying stress to the cooled and programmed form of the SMP for a temporary deformation. As a further step, heat as a stimulus is introduced in the process. In this case, the given temperature might be lower than T_p but higher than T_g, due to which the shape recovery will occur, and the SMP will revert to its original form.

3. Low-cost shape memory property characterization method

For the widespread use and application of FDM 4D printing, it is highly necessary to find simple and economically feasible solutions

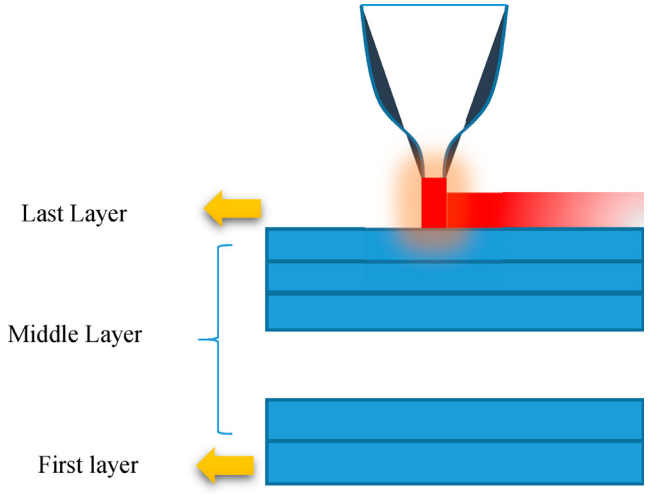


Fig. 4. Programming of SMP during FDM 4D printing [24].

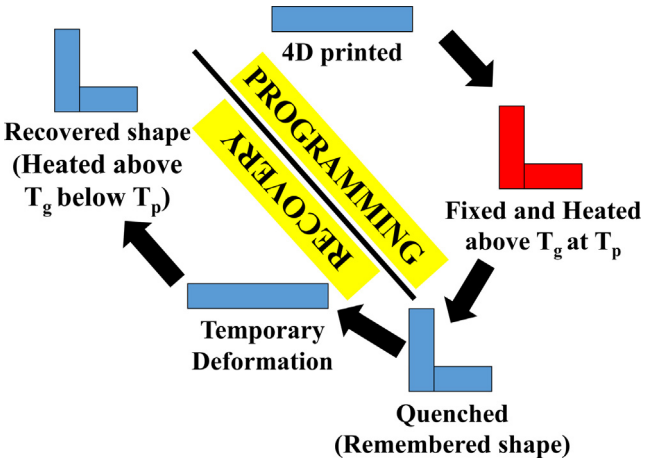


Fig. 5. Programming of SMP after printing by loading-annealing-quenching.

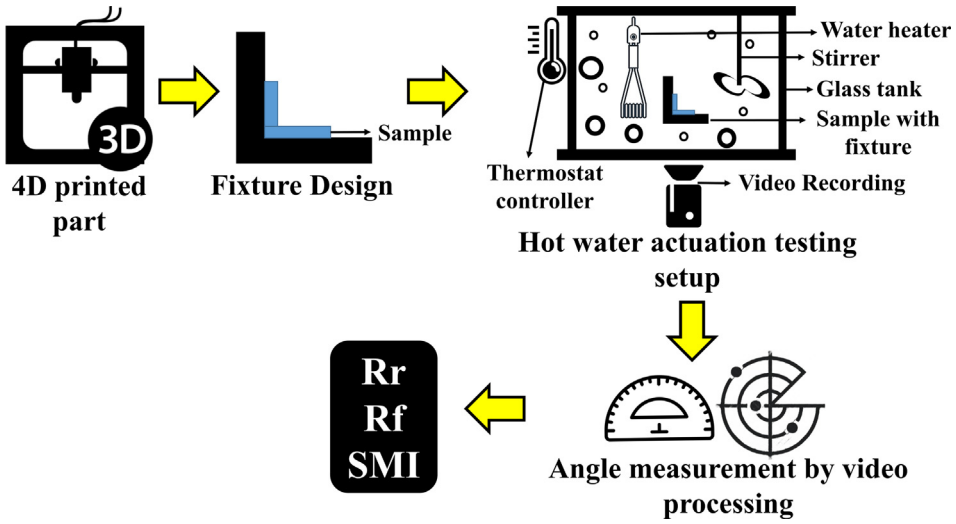


Fig. 6. Low-cost shape memory property characterization process for FDM 4D printed parts.

to program and characterize the shape memory properties. The cost of FDM technology has been reducing at an exponential rate due to the replication rapid prototype (RepRap) and do-it-yourself (DIY) culture. In the present scenario, a fully assembled FDM 3D printer can cost as low as $100. This has opened doors for hobbyists, children, and low-income countries to start using FDM 3D printers for rapid prototyping and other various application. Therefore, using FDM technology, low-cost SMP material, economic programming, and shape memory characterizations can be utilized for the realization of low-cost 4D printing technology. This can have a huge impact as anyone will be able to create shape-morphing structures easily. As FDM technology is the only AM process that has been used even in space. Therefore, the findings on low-cost FDM 4D printing technology can also be used to create shape-changing structures directly in space for deployable self-morphing applications. The complete process has been depicted in Fig. 6.

3.1. Folding/ bending angle measurement characterization

Bending angle measurement characterization has been utilized by various researchers worldwide for the shape memory behavior study of both shape memory polymers and shape memory alloys [25 26 27 28 29]. It is an economical yet robust method of shape memory characterization. The process of characterization starts with the 4D printed part that has been removed from the build platform after the printing is over. The 4D printed samples are then fixed in a fixture that is pre-designed in a way that the user wants to make the sample remember its shape (in case of loading-annealing-quenching type programming). In the case of programming due to pre-strain during the FDM process, the fixture is designed to provide a temporary deformation. Once it is completed, then the hot water actuation testing setup is used for the thermo-responsive actuation and also for programming by annealing. The video data is collected during the actuation. This data can be used by any open-source video analysis tool, and the accurate angle, acceleration, actuation time, and speed can be calculated. One of the video analysis and modeling software is "Tracker" [30]. Once the variables have been calculated by the video analysis software, different equations can be used to calculate various shape memory properties like shape recovery ratio (R_r), shape fixation ratio (R_f), and shape memory index (SMI). An elastic rubber band, which is commonly used in our homes,when stretched and released, the recovery is 100 %. There is, however, no fixation or memory effect. As a result, the elastic rubber band cannot be considered a shape memory material (SMM) [31]. For a material to be the SMM, it must have both shape recovery and shape fixation.SMI assesses the effectiveness of the overall SME in an SMM by measuring the combined effect of the two critical ratios, shape recovery ratio and shape fixation ratio. The shape recovery ratio assesses an SMM's ability to return to its original shape after being temporarily fixed. The shape fixity ratio assesses an SMM's ability to be fixed to a user-specified temporary or intermediate shape [32]. SMI assesses the effectiveness of the overall SME in an SMP by measuring the combined effect of the two critical ratios, shape recovery ratio and shape fixation ratio.

3.1.1. Fixture design for programming and temporary deformation

The fixture design is an important aspect of the programming as it will help to set the programmed shape, and it can also be used for providing a temporary deformation. An example of the fixture design for 90 degrees bending actuator for FDM 4D printed SMP mounted on it has been shown in Fig. 7. It should be made up of a material that will have heat resistance and will not deform at round T_p of the SMP. Wood and metals are good choices for the fixture material. Further fabrication of the fixture can be done using conventional manufacturing processes.

3.2. Setup for hot water actuation testing and programming by annealing

A low-cost setup is highly necessary for calculating the shape memory properties and also to provide the preheat temperature for temporary deformation. The setup has to be made such that it can also be used for programming by annealing the samples. An in-house build setup has been developed successfully at the Direct Digital Manufacturing Laboratory, Indian Institute of Technology, Kharagpur, WB, India (Fig. 8). The setup consists of a glass chamber with fully filled deionized water. It consists of a water heater in the form of an immersion rod for heating the distilled water. There is a mechanical stirrer that helps distribute the heat uniformly throughout the chamber. The temperature probe shows the actual temperature inside the chamber. The probe used is a negative temperature coefficient (NTC) of 10,000 Ω, a waterproof thermistor temperature sensor. A thermostat controller based on the relay circuit has been used for the ON/OFF switch to control the temperature to the nearest value of the set temperature. It is a DC 12 V digital temperature controller with a thermostat switch

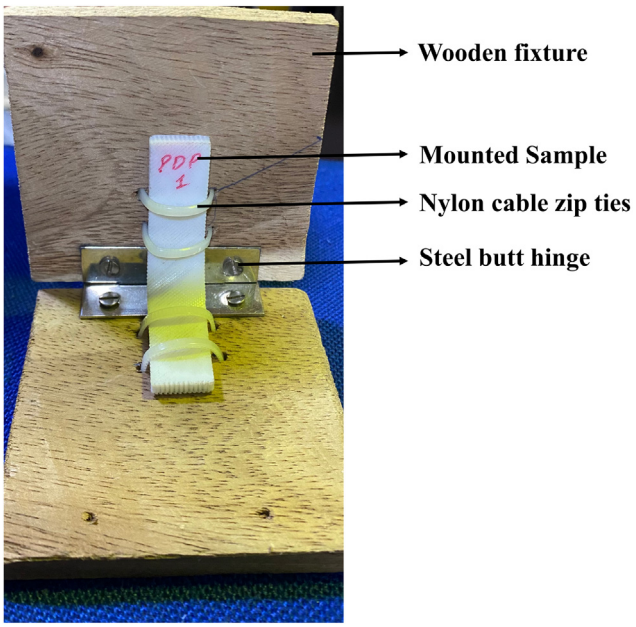


Fig. 7. An example of fixture design for 90-degree programming and recovery characterization of bending actuator.

(Model: W3230). The temperature control range is from 55 degreesC to 120 degreesC with a resolution, control accuracy, and measurement accuracy of 0.1 degreesC. The working environment range of the instrument is 10–60 degreesC temperature and 20 %-85 % humidity.

3.3. Various related angles and their measurement for different programming methods

Angle measurement schemes are economical ways of finding out the shape memory properties with high accuracy. The angle measurement schemes can vary depending upon the type of 4D printed actuator, as shown in Fig. 9 (bending or folding). For both kinds of actuators, the programming can be done in two different ways, as explained in section 2.2.1 and section 2.2.2. The first case considered is for programming due to pre-strain during the FDM process. Using this programming concept and folding actuator (Fig. 9 (a)), the printed shape is equal to the programmed shape. Therefore, the maximum programmed angle is 180 degrees. As it is a well-known fact that FDM 3D printing has got certain dimensional inaccuracy, the actual programmed (also called fixed) angle is ∠fpd, and the fixation error ∠dpe. A fixture is used to provide a temporary deformation. The required deformation is 90 degrees and hence ∠fpa, but due to an error in deformation, the actual deformation angle is ∠fpb by considering the deformation error of ∠bpa. After the recovery cycle, the final position is marked, and the angle is measured. The angle is ∠bpc. Now by dividing the fixed angle ∠fpd by the maximum programmed angle, i.e., 180 degrees, the fixation ratio R_f is obtained. The recovery ratio, R_r can be calculated by subtracting the angle ∠bpc by ∠fpd and then dividing by ∠bpc. The SMI is the combined effect of R_f and R_r.

Similarly, for the programming by loading-annealing-quenching post-process, the printed angle and the programmed angle may have variations. In that case, the printed angle is ∠fpe (180 degrees). Let's assume that it has to be programmed to 90 degrees. Now the programmed angle should be ∠ape, but due to the errors during the programming, the actual fixed angle is ∠bpe

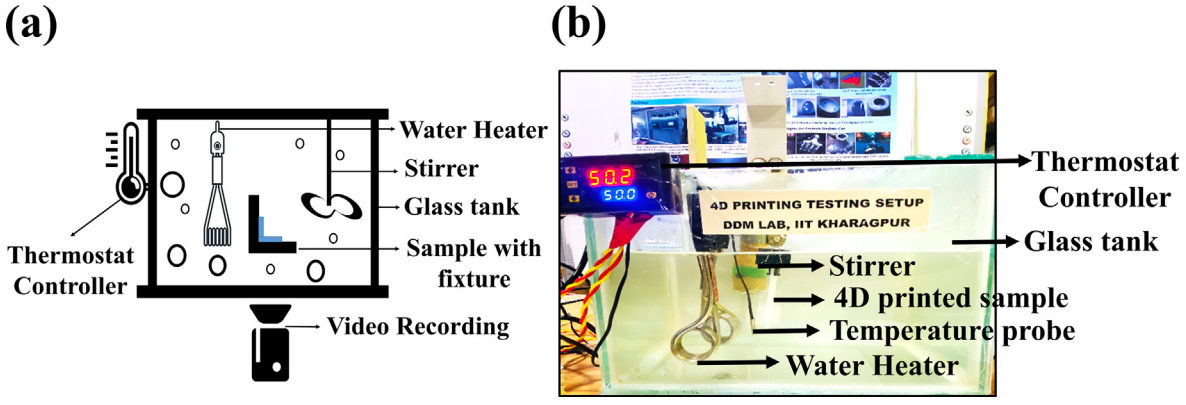


Fig. 8. Hot water shape recovering and annealing programming characterization apparatus (a) Schematic (b) Actual.

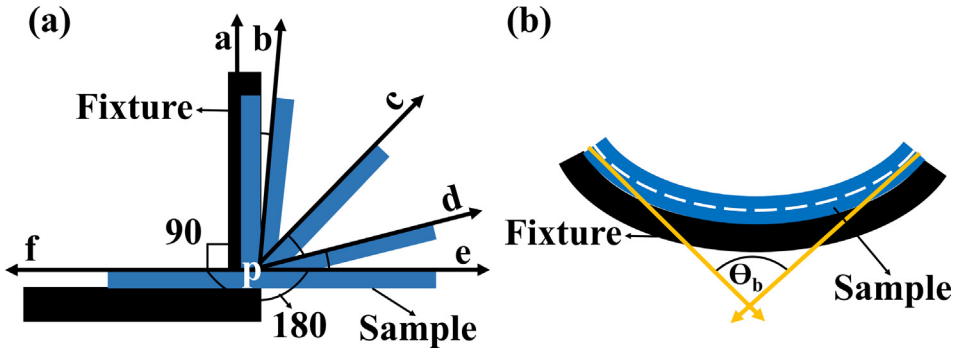


Fig. 9. Angle measurement methods (a) Folding actuation (b) Bending actuation.

by considering the fixation error of ∠apb. Therefore, R_f is ∠bpe divided by 90 degrees. After a temporary deformation (180 degrees) is provided, by considering the deformation error of ∠dpe, the angle of actual deformation is ∠apd. After the shape recovery, the position is marked, and the angle measured is ∠dpc. Therefore, R_r is ∠dpc subtracted by ∠bpe and then divided by ∠dpc.

For the case of bending actuator, for both the cases of programming strategies, the variation in the angle θ_b can be marked for both programming and recovery, and the values of R_f, R_r, and SMI can be easily computed.

4. Conclusion

4D printing technology is a revolutionary development that has the potential to transform the way we think about manufacturing. The biggest disadvantage of any new technology is its high cost. To reduce this, there needs to be a concerted effort from the engineering and research sectors to improve its efficiency and effectiveness. This can be done by combining existing economically sustainable technologies along with creating new methodologies such as low-cost FDM 3D printing, SMP materials, and characterization methods. This way, 4D printing technology can be adopted by a larger number of people, thus allowing the development of more applications, products, and services in different fields such as healthcare, manufacturing, and automotive to enhance existing products and services. The present study discusses in detail the types of 4D printing and the mechanism of the shape memory effect in TR-SMP. Various programming strategies and low-cost characterization techniques using the in-house build setup and angle measurement schemes have been presented. These results will be helpful for future researchers to gain knowledge about 4D printing technology and start various DIY low-cost 4D printing projects. These results can also be utilized further to create structures that will be self-morphing and reconfigurable.

CRediT authorship contribution statement

Bijaya Bikram Samal: Conceptualization, Methodology, Investigation, Formal analysis, Validation, Writing - original draft. **Anita Jena:** Formal analysis, Visualization, Writing - review & editing. **Shailendra Kumar Varshney:** Supervision, Resources, Writing - review & editing. **Cheruvu Siva Kumar:** Supervision, Resources, Writing - review & editing.

Data availability

No data was used for the research described in the article.

Declaration of Competing Interest

The authors declare that no financial interests/personal relationships that may have appeared to influence the work reported in this paper: Bijaya Bikram Samal reports was provided by Indian Institute of Technology Kharagpur.

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